

● HARD

● PROBLEM-SOLVING AND DATA ANALYSIS

Percentages

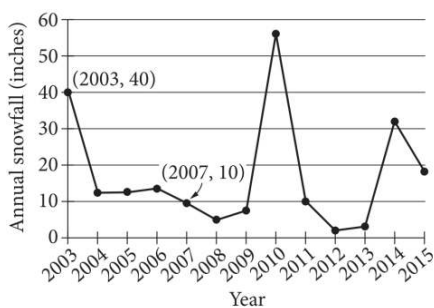
10 questions · ~15 min

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Q1

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS



The line graph shows the total amount of snow, in inches, recorded each year in Washington, DC, from 2003 to 2015. If $p\%$ is the percent decrease in the annual snowfall from 2003 to 2007, what is the value of p ?

STUDENT-PRODUCED RESPONSE

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q2

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

The regular price of a shirt at a store is \$ 11.70. The sale price of the shirt is 80% less than the regular price, and the sale price is 30% greater than the store's cost for the shirt. What was the store's cost, in dollars, for the shirt? (Disregard the \$ sign when entering your answer. For example, if your answer is \$ 4.97, enter 4.97)

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Year	Subscriptions sold
2012	5,600
2013	5,880

The manager of an online news service received the report above on the number of subscriptions sold by the service. The manager estimated that the percent increase from 2012 to 2013 would be double the percent increase from 2013 to 2014. How many subscriptions did the manager expect would be sold in 2014?

- A. 6,020
- B. 6,027
- C. 6,440
- D. 6,468

Q4

PROBLEM-SOLVING AND DATA ANALYSIS

The expression $0.35x$ represents the result of decreasing a positive quantity x by what percent?

- A. 3.5%
- B. 35%
- C. 6.5%
- D. 65%

The number a is 190% greater than the number b . The number b is 80% less than 24. What is the value of a ?

- A. 9.12
- B. 13.92
- C. 26.40
- D. 36.48

Q6

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

The number a is 70% less than the positive number b . The number c is 60% greater than a . The number c is how many times b ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

PROBLEM-SOLVING AND DATA ANALYSIS

STUDENT-PRODUCED RESPONSE

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

8 / 11

After 20% of the original number of marbles in a group were removed from the group, 360 marbles remained in the group. How many marbles were removed from the group?

- A. 72
- B. 90
- C. 450
- D. 1,800

Q9

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

The number a is 110% greater than the number b . The number b is 90% less than 47. What is the value of a ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

37% of the items in a box are green. Of those, 37% are also rectangular. Of the green rectangular items, 42% are also metal. Which of the following is closest to the percentage of the items in the box that are not rectangular green metal items?

- A. 1.16%
- B. 57.50%
- C. 94.25%
- D. 98.84%